



■ CS177 Bioinformatics: Software Development and Applications

Lecture 23: AI Virtual Cell (AIVC)

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Learning objectives



- At the end of this lecture, you should be able to:
 - Understand the basics of whole-cell modeling (aims, methodology, and challenges)
 - Describe how to model **cell cycle control system**
 - Explain the key difference between ~~whole-cell modeling~~ and AI virtual cell (AIVC)
 - Define and explain the concept of **universal representation (UR)** at multiple scales in molecular and cell biology
 - Describe the **virtual instruments (VIs)**, including the manipulators and decoders
 - Give a few examples of applying AIVC, e.g. drug discovery, cell engineering, cancer medicine, scientific research





Whole-Cell Modeling



Aims of whole-cell modeling

- Whole-cell models aim to
 - Represent **all of the chemical reactions in a cell** and all of the physical processes that influence their rates
 - Predict the **behavioral trajectories** of single cells over their life cycles, with each simulation representing a different cell within a heterogeneous clonal population

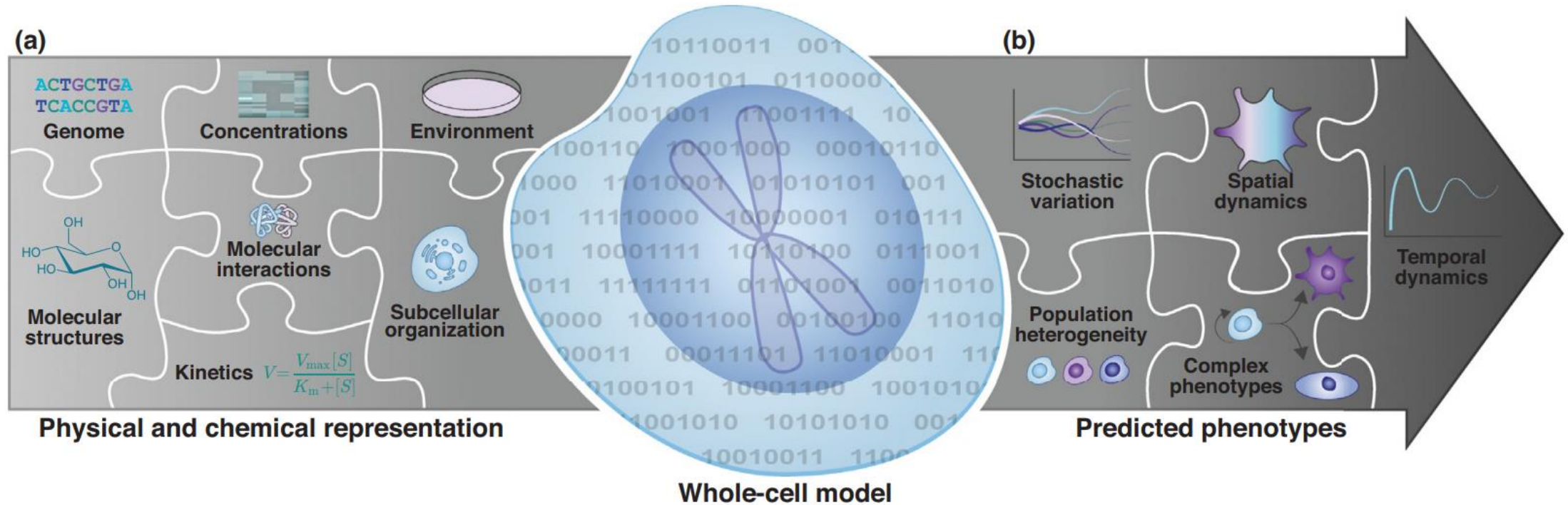
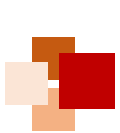


Fig. 1 of Goldberg et al. (2018) Current Opinion in Biotechnology, 51:97–102



Whole-cell modeling methodology



- Recent advances in data aggregation, model design, model representation, and simulation are also rapidly making whole-cell modeling feasible. Emerging methods and tools include:
 - Data aggregation and organization
 - Scalable model design
 - Model languages
 - Simulation, calibration and verification
 - Simulation results analysis

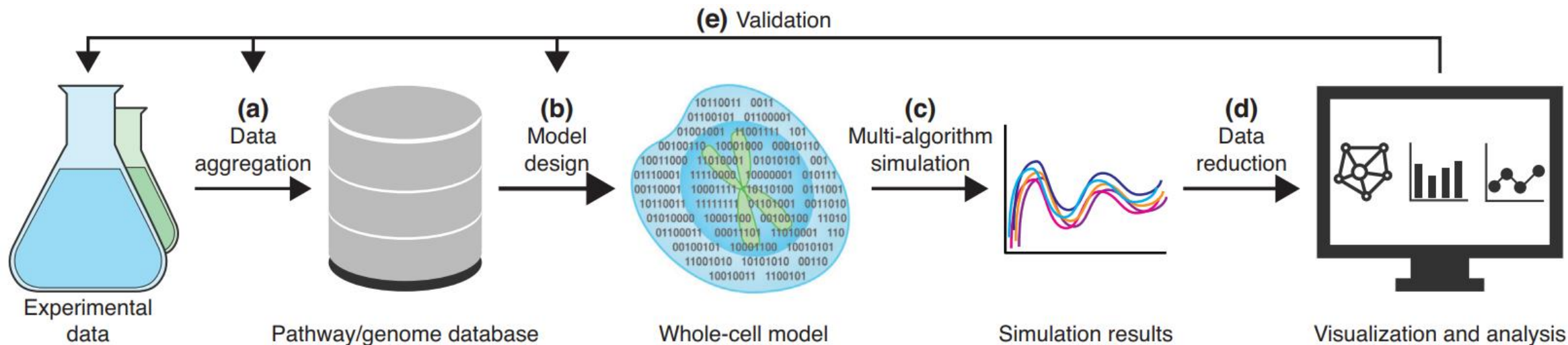


Fig. 2 of Goldberg et al. (2018) Current Opinion in Biotechnology, 51:97–102

Technical challenges



- A recent survey of the bio-modeling community (Szigeti et al. 2018) shows the following **bottlenecks to whole-cell modeling**:
 - Inadequate experimental methods and data repositories
 - Incompatibility among published models
 - Inadequate software tools for designing, calibrating and validating models
 - Lack of adequate formats for representing models

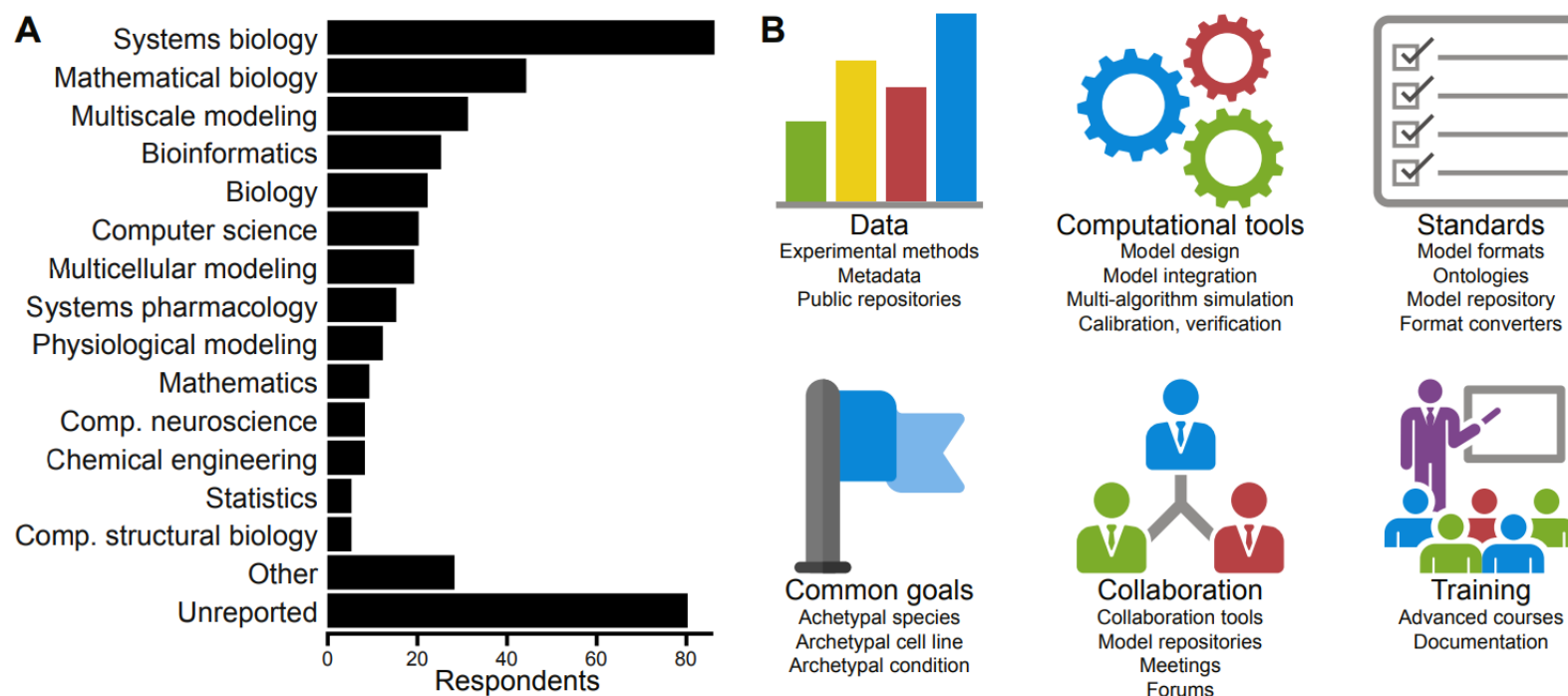


Fig. 2 of Szigeti et al. (2018) Current Opinion in Systems Biology, 7:8–15



In Silico Simulation of Cell Cycle Control

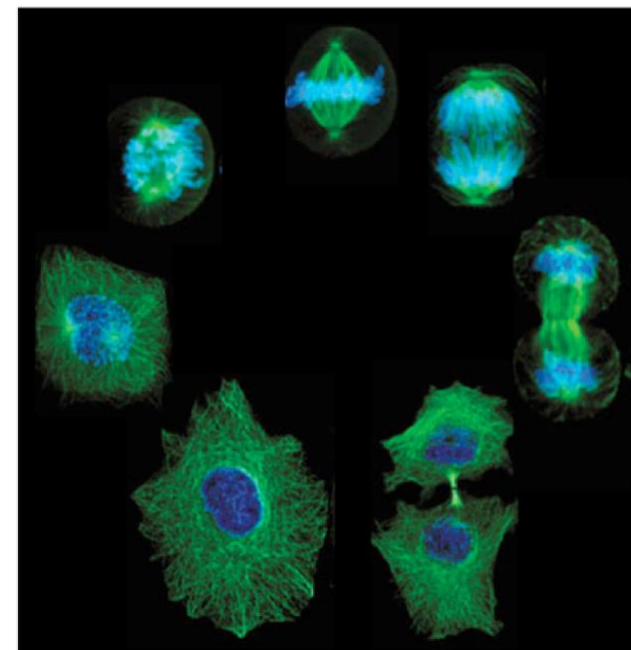




The cell cycle

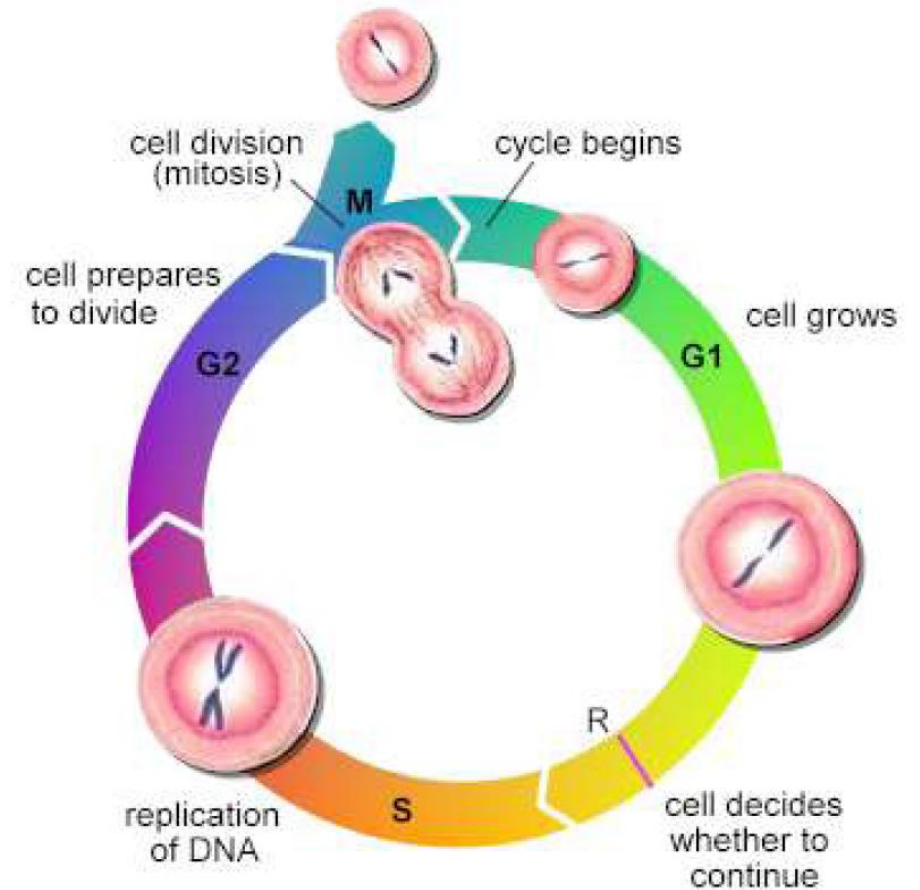


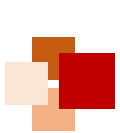
- Among the cellular processes, the cell division cycle is one of the most fascinating and well studied, due to
 - The fine regulation mechanisms used for the synchronization and proof checking of the process
 - The relevance for understanding the occurrence of many diseases, such as cancer



The cell cycle (cont' d)

- Each cell reproduces by performing an orderly sequence of events
- The classical model of cell cycle comprises 4 sequential phases:
 - **S (synthesis) phase**: The DNA is replicated to produce two identical copies
 - **G2 (gap) phase**: New proteins are synthesized and cell approximately doubles in size
 - **M (mitosis) phase**: Matching chromosomes are pulled to opposite poles of the cell and cytokinesis pinches the cell in half
 - **G1 (gap) phase**: Each daughter cell replicates again after a gap period, or enters a quiescent phase (G0)

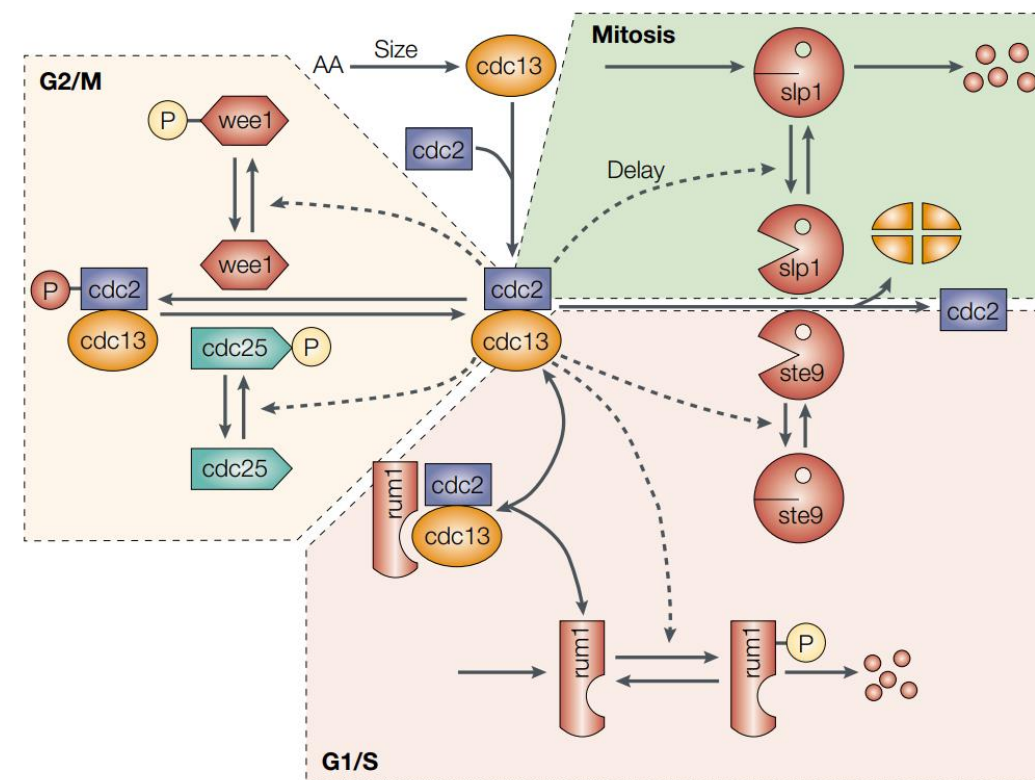




Cell cycle control system



- In the sequel, we will follow the presentation given in
Tyson, Chen, Novak, Network Dynamics and Cell Physiology, Nature Reviews
Molecular Cell Biology, 2:908-916, 2001
- The scheme represents a wiring diagram of the basic mechanisms involved in the regulation of the cell cycle
- Each node represents a protein (or complex of proteins)
- Solid arrows represent the biochemical process that produce, degrade, activate or inactivate the attached proteins
- Dashed arrows represent the regulatory influence of one protein on a biochemical process



The cell-cycle control system in fission yeast



Cell cycle in mammalian cells

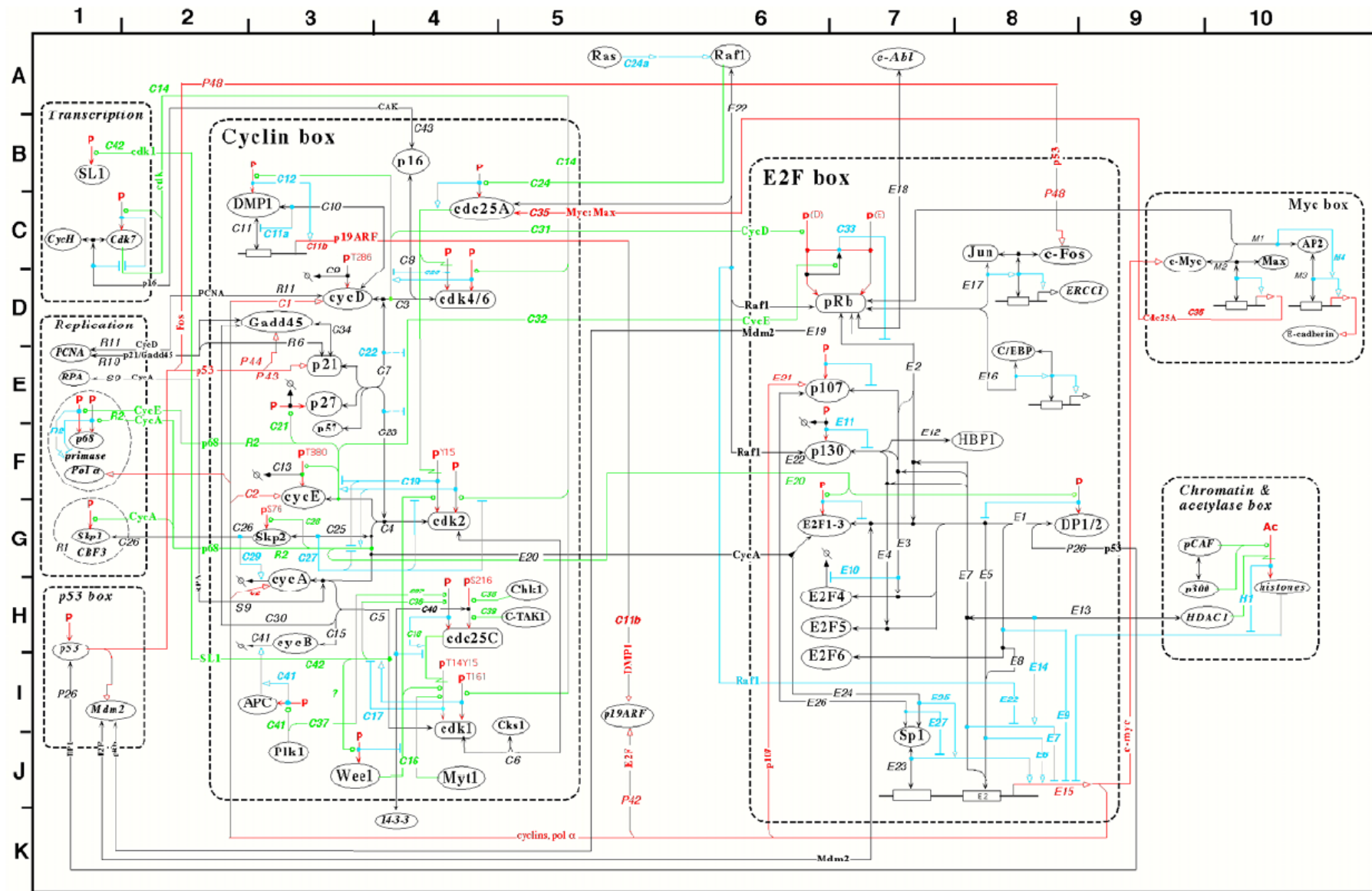


Figure 6A: The Cyclin - E2F cell cycle control system (version 3a - June 8, 1999)

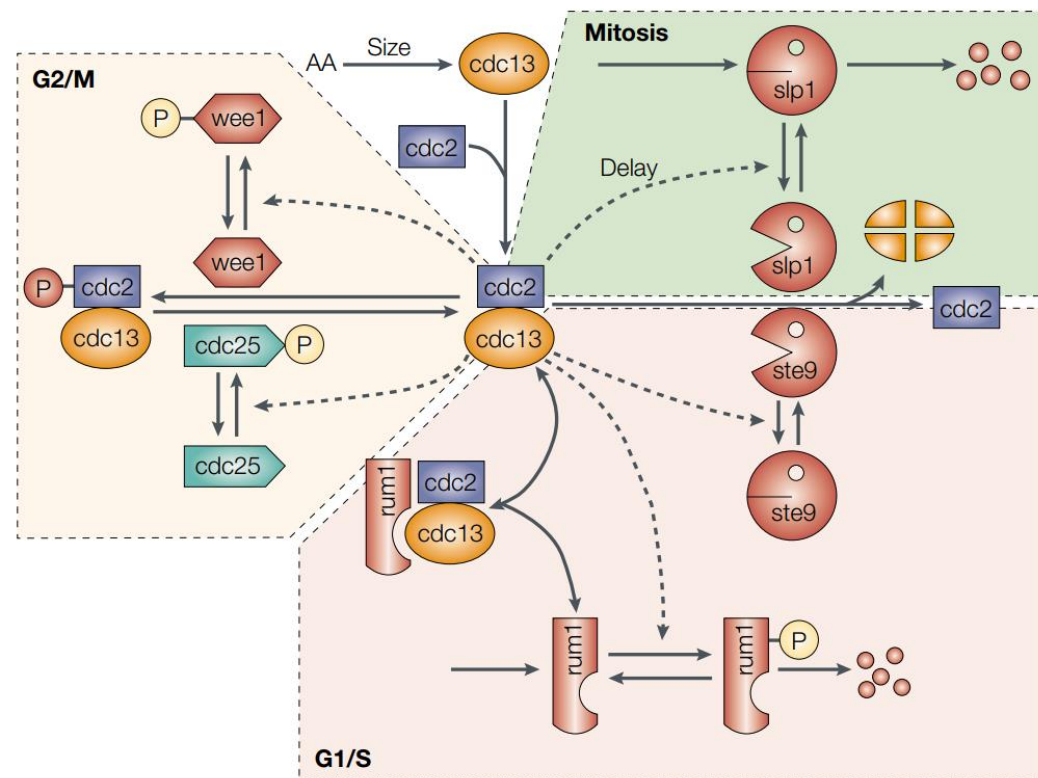
Model derivation

- The ODE model can be derived as:

动力学参数

$$\frac{d}{dt}[\text{cdc2-cdc13}] = k_1 \cdot \text{size} - k_2[\text{cdc2-cdc13}] - k_3[\text{rum1}][\text{cdc2-cdc13}] + k_4[\text{rum1-cdc2-cdc13}] - k_5[\text{cdc2-cdc13}] + k_6[\text{cdc2P-cdc13}]$$

- The kinetic parameters are, in turn, determined by the activity of the other species
- For instance, the parameter k_2 quantifies the degradation rate of **cdc13**, which depends on the activities of **slp1** and **ste9**



Model equations



$$\frac{d[\text{Cdc13}_T]}{dt} = k_1 M - (k'_2 + k''_2[\text{Ste9}] + k'''_2[\text{Slp1}])[\text{Cdc13}_T], \quad (1)$$

$$\begin{aligned} \frac{d[\text{preMPF}]}{dt} = & k_{\text{wee}}([\text{Cdc13}_T] - [\text{preMPF}]) - k_{25}[\text{preMPF}] - (k'_2 \\ & + k''_2[\text{Ste9}] + k'''_2[\text{Slp1}])[\text{preMPF}], \end{aligned} \quad (2)$$

$$\begin{aligned} \frac{d[\text{Ste9}]}{dt} = & (k'_3 + k''_3[\text{Slp1}]) \frac{1 - [\text{Ste9}]}{J_3 + 1 - [\text{Ste9}]} - (k'_4[\text{SK}] \\ & + k_4[\text{MPF}]) \frac{[\text{Ste9}]}{J_4 + [\text{Ste9}]}, \end{aligned} \quad (3)$$

$$\frac{d[\text{Slp1}_T]}{dt} = k'_5 + k''_5 \frac{[\text{MPF}]^4}{J_5^4 + [\text{MPF}]^4} - k_6[\text{Slp1}_T], \quad (4)$$

$$\begin{aligned} \frac{d[\text{Slp1}]}{dt} = & k_7[\text{IEP}] \frac{[\text{Slp1}_T] - [\text{Slp1}]}{J_7 + [\text{Slp1}_T] - [\text{Slp1}]} \\ & - k_8 \frac{[\text{Slp1}]}{J_8 + [\text{Slp1}]} - k_6[\text{Slp1}], \end{aligned} \quad (5)$$

$$k_{\text{wee}} = k'_{\text{wee}} + (k''_{\text{wee}} - k'_{\text{wee}}) G(V_{\text{awee}}, V_{\text{iwee}}[\text{MPF}], J_{\text{awee}}, J_{\text{iwee}}),$$

$$k_{25} = k'_{25} + (k''_{25} - k'_{25}) G(V_{\text{a25}}[\text{MPF}], V_{\text{i25}}, J_{\text{a25}}, J_{\text{i25}}),$$

$$\Sigma = [\text{Cdc13}_T] + [\text{Rum1}_T] + K_{\text{diss}},$$

$$G(a, b, c, d) = \frac{2ad}{b - a + bc + ad + \sqrt{(b - a + bc + ad)^2 - 4ad(b - a)}}.$$

$$\frac{d[\text{IEP}]}{dt} = k_9[\text{MPF}] \frac{1 - [\text{IEP}]}{J_9 + 1 - [\text{IEP}]} - k_{10} \frac{[\text{IEP}]}{J_{10} + [\text{IEP}]}, \quad (6)$$

$$\frac{d[\text{Rum1}_T]}{dt} = k_{11} - (k_{12} + k'_{12}[\text{SK}] + k''_{12}[\text{MPF}])[\text{Rum1}_T], \quad (7)$$

$$\frac{d[\text{SK}]}{dt} = k_{13}[\text{TF}] - k_{14}[\text{SK}], \quad (8)$$

$$\frac{dM}{dt} = \mu M, \quad (9)$$

ODE

$$[\text{Trimer}] = \frac{2[\text{Cdc13}_T][\text{Rum1}_T]}{\Sigma + \sqrt{\Sigma^2 - 4[\text{Cdc13}_T][\text{Rum1}_T]}}, \quad (10)$$

$$[\text{MPF}] = \frac{([\text{Cdc13}_T] - [\text{preMPF}])([\text{Cdc13}_T] - [\text{Trimer}])}{[\text{Cdc13}_T]}, \quad (11)$$

$$[\text{TF}] = G(k_{15}M, k'_{16} + k''_{16}[\text{MPF}], J_{15}, J_{16}), \quad (12)$$

- 9 ODEs
- 3 algebraic equations
- ~30 parameters

Novak et al. Chaos 11(1):277-286, 2001



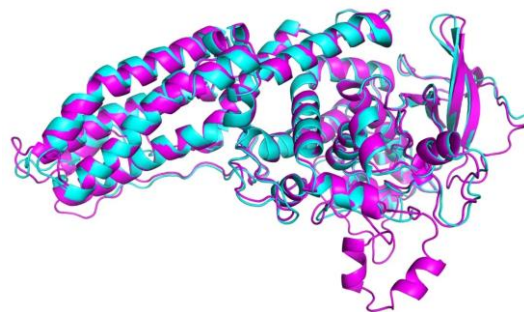
AI Virtual Cell (AIVC)



AI for Life Science: from protein to cell

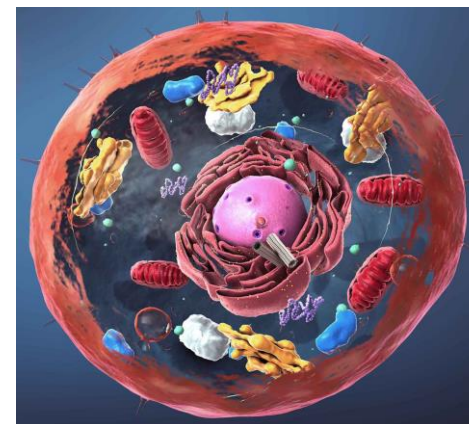


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AlphaFold

Structural Biology

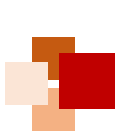


AI Virtual Cell (AIVC)

Systems Biology



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From Whole-Cell Modeling to AIVC



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Cell 2012 (Stanford)

Cell 2024 (Stanford)

Theory

A Whole-Cell Computational Model Predicts Phenotype from Genotype

Jonathan R. Karr,^{1,4} Jayodita C. Sanghvi,^{2,4} Derek N. Macklin,² Miriam V. Gutschow,² Jared M. Jacobs,² Benjamin Bolival, Jr.,² Nacyra Assad-Garcia,³ John I. Glass,³ and Markus W. Covert^{2,*}

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²Department of Bioengineering

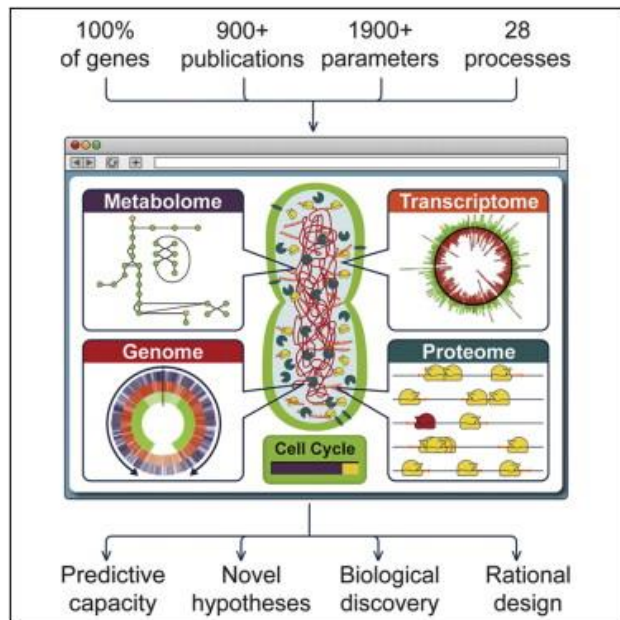
Stanford University, Stanford, CA 94305, USA

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⁴These authors contributed equally to this work

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<http://dx.doi.org/10.1016/j.cell.2012.05.044>



Cell

Cell

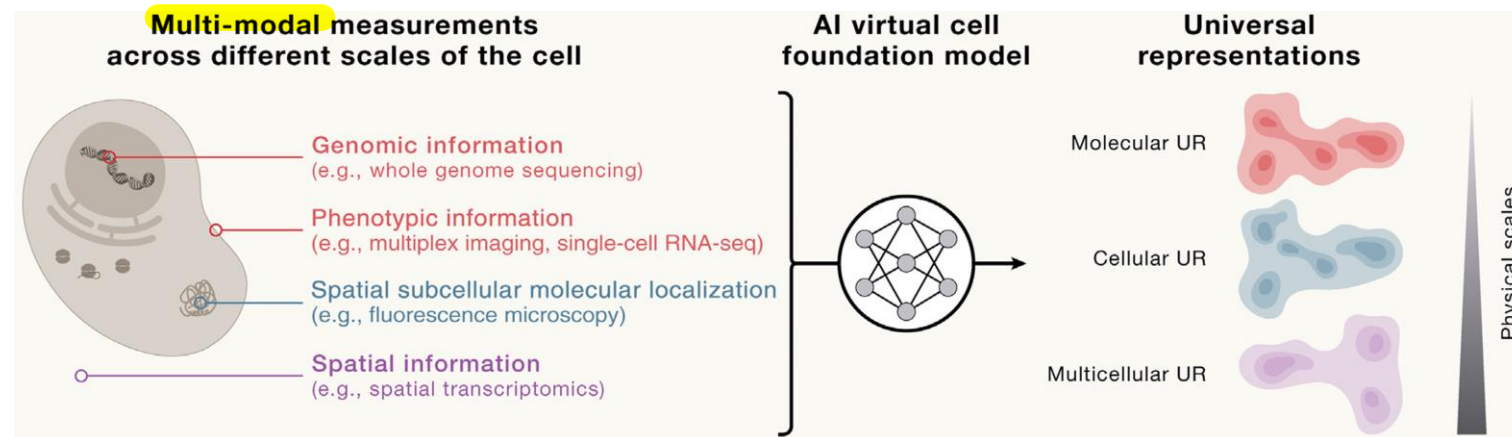
Leading Edge

CellPress
OPEN ACCESS

Perspective

How to build the virtual cell with artificial intelligence: Priorities and opportunities

Charlotte Bunne,^{1,2,3,4,50} Yusuf Roohani,^{1,3,5,50} Yanay Rosen,^{1,3,50} Ankit Gupta,^{3,6} Xikun Zhang,^{1,3,7} Marcel Roed,^{1,3} Theo Alexandrov,^{8,9} Mohammed AlQuraishi,⁹ Patricia Brennan,³ Daniel B. Burkhardt,¹¹ Andrea Califano,^{10,12,13} Jonah Cool,³ Abby F. Dernburg,¹⁴ Kirsty Ewing,³ Emily B. Fox,^{1,15,16} Matthias Haury,¹⁷ Amy E. Herr,^{16,18} Eric Horvitz,¹⁹ Patrick D. Hsu,^{5,18,20} Viren Jain,²¹ Gregory R. Johnson,²² Thomas Kalil,²³ David R. Kelley,²⁴ Shana O. Kelley,^{25,26} Anna Kreshuk,²⁷ Tim Mitchison,²⁸ Stephani Otte,¹⁷ Jay Shendure,^{29,30,31,32} Nicholas J. Sofroniew,³³ Fabian Theis,^{34,35,36} Christina V. Theodoris,^{37,38} Srigokul Upadhyayula,^{14,16,39} Marc Valer,³ Bo Wang,^{40,41} Eric Xing,^{42,43} Serena Yeung-Levy,^{1,44} Marinka Zitnik,^{45,46,47} Theofanis Karaletsos,^{3,*} Aviv Regev,^{2,*} Emma Lundberg,^{3,6,7,48,*} Jure Leskovec,^{1,3,*} and Stephen R. Quake^{3,7,49,*}



Fundamental difference:

- Whole-cell modeling is knowledge (or rule)-driven
- AI virtual cell is mainly data-driven

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Challenges to rule-based cell modeling



- Existing rule-based cell models often combine assumptions about the underlying biological mechanisms with parameters fit from data, and use explicitly defined mathematical or computational approaches, e.g.:
 - Differential equations
 - Stochastic simulations
 - Agent-based models
- Challenges to the rule-based cell models:
 - Multi-scale modeling: Cells operate on multiple scales across both time and space
 - Diverse processes with massive numbers of interacting components: E.g. gene regulation, metabolic pathways, and signal transduction
 - Nonlinear dynamics: small changes in inputs can lead to complex changes in outputs





Two revolutions enable data-driven cell modeling

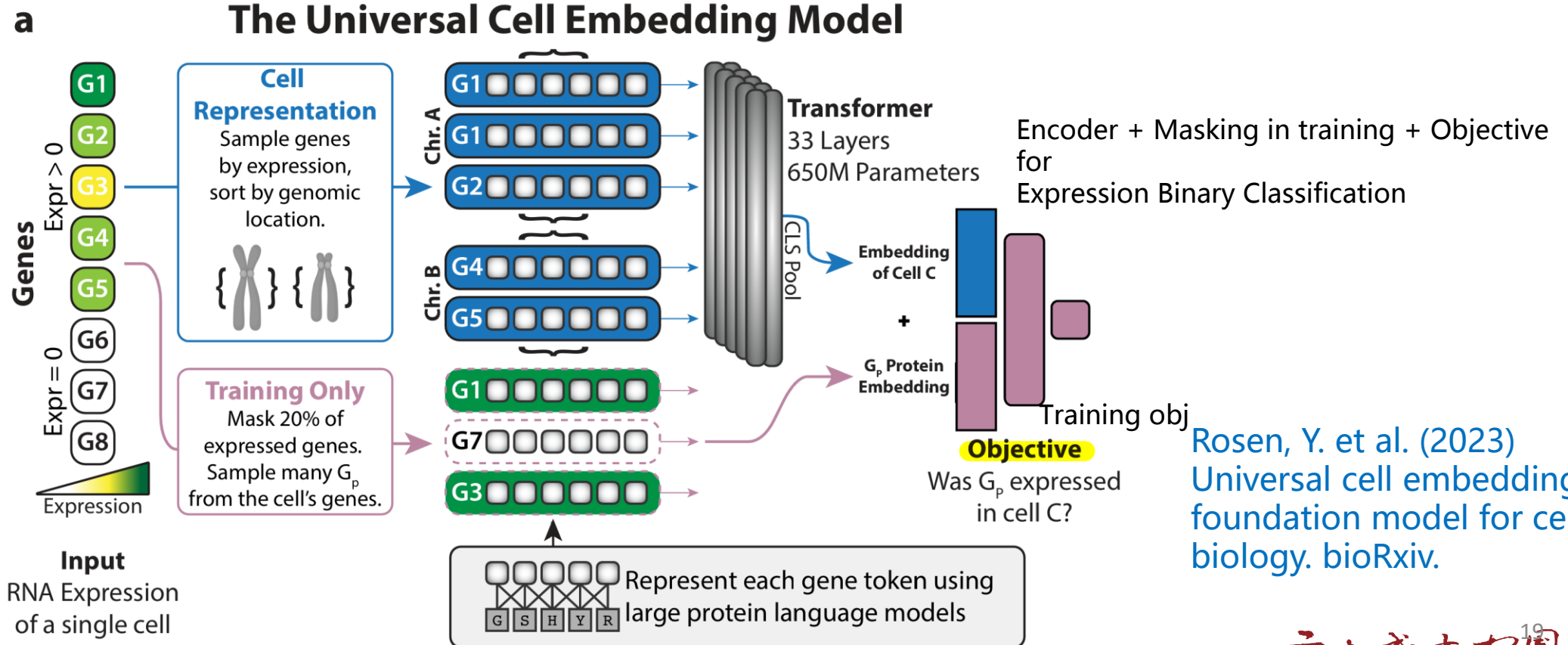


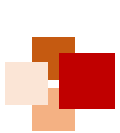
- **Revolution 1 (experimental): Omics data**
 - Increase in the high-throughput measurement technologies has led to the collection of large and growing datasets
 - Data doubling every 6 months for the past several years
 - The ability to couple the measurements with systematic perturbations
- **Revolution 2 (computational): AI technologies**
 - Enhanced ability to learn patterns and processes directly from data
 - Without needing explicit rules or human annotation
 - E.g. AlphaFold and protein language models
 - Predictive, generative, and queryable
- **An ambitious vision of AIVC:** A multi-scale, multi-modal, large-neural-network-based, fully data-driven model
 - Enable fast-paced in silico studies
 - Bridges between computational methods and confirmatory wet-lab experiments



Universal representation (UR)

- **Definition:** Universal Representation (UR) is a multi-scale, multi-modal embedding that captures the **state and behavior of biological entities** across different contexts and modalities.
- **Purpose:** To create a unified framework for representing and analyzing biological data, enabling high-fidelity simulations and hypothesis generation.



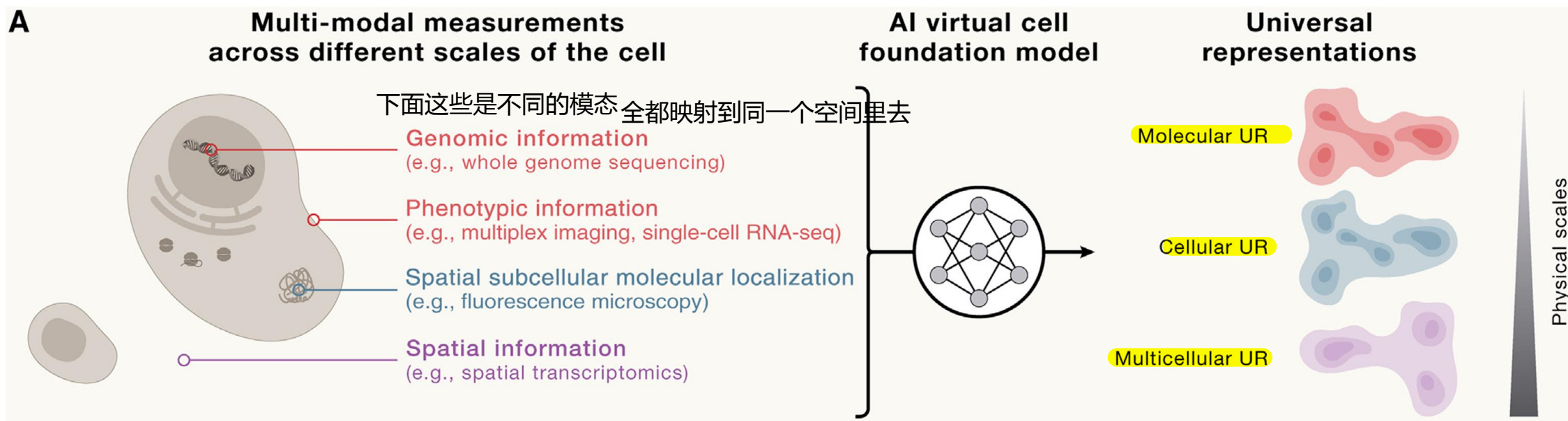


Features of universal representation (UR)



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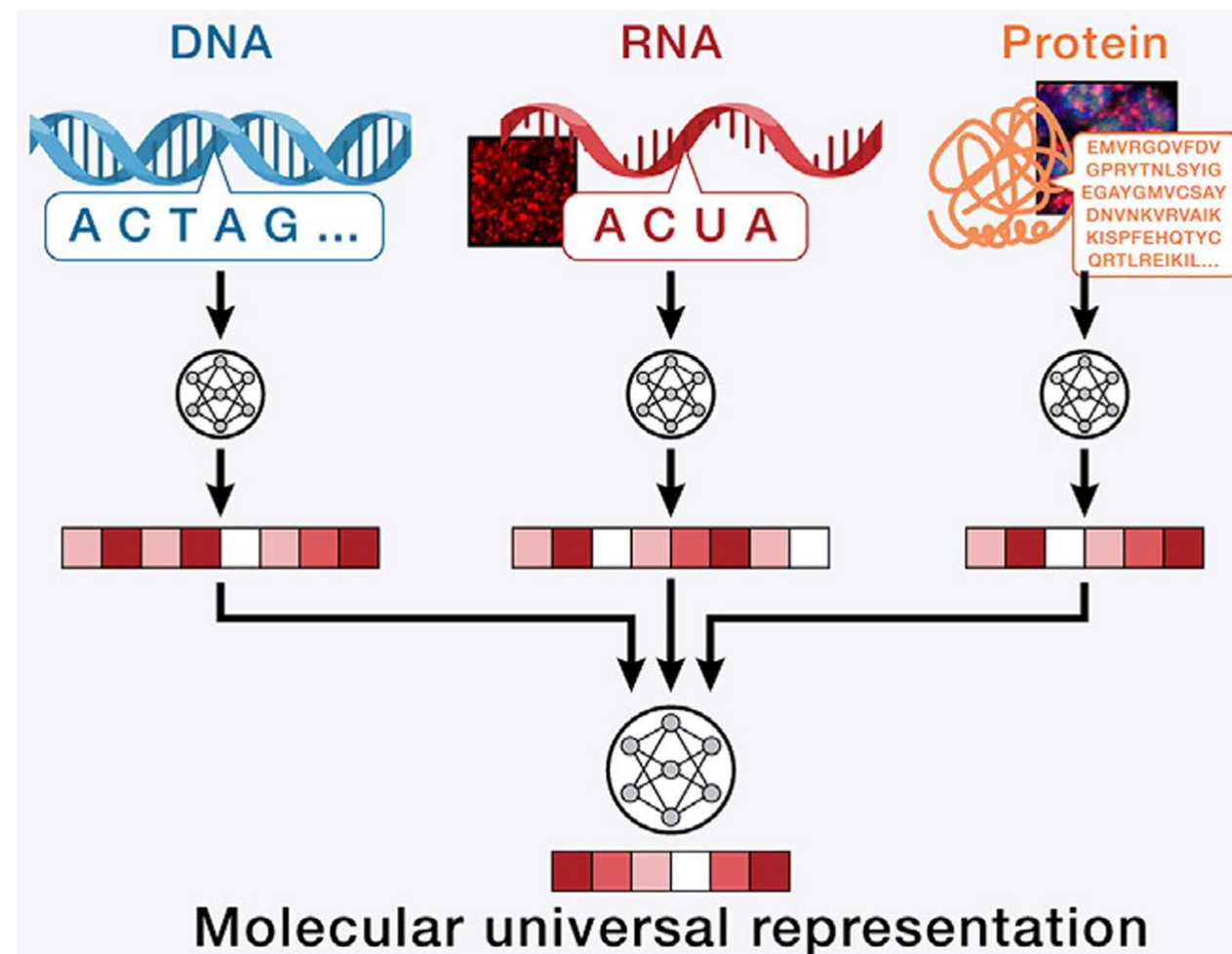
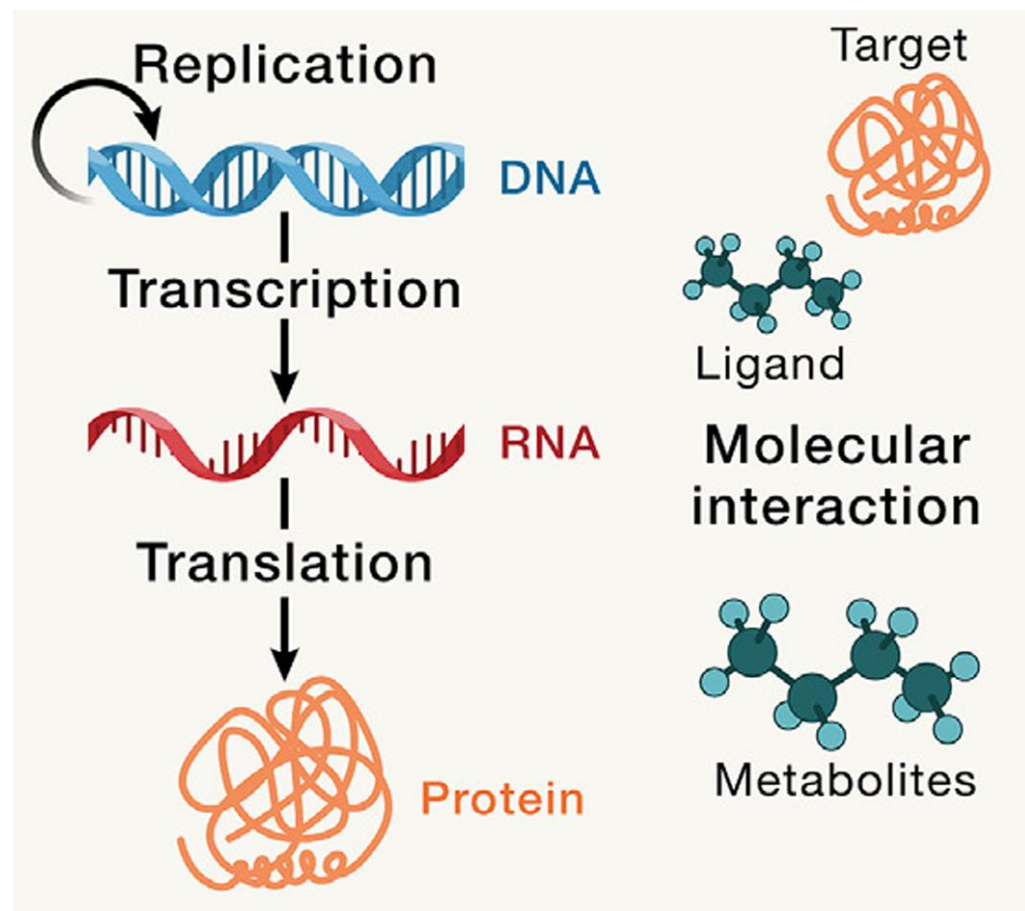
- **Multi-Scale Integration:** UR integrates data across molecular, cellular, and multicellular levels
 - **Molecular Scale:** Represents individual molecules (DNA, RNA, proteins) and their interactions
 - **Cellular Scale:** Captures the state of individual cells, including gene expression, protein levels, and spatial organization
 - **Multicellular Scale:** Models interactions between cells and their organization into tissues and organs
- **Multi-Modal Data:** UR can be generated from various data types, including sequencing, imaging, and proteomics



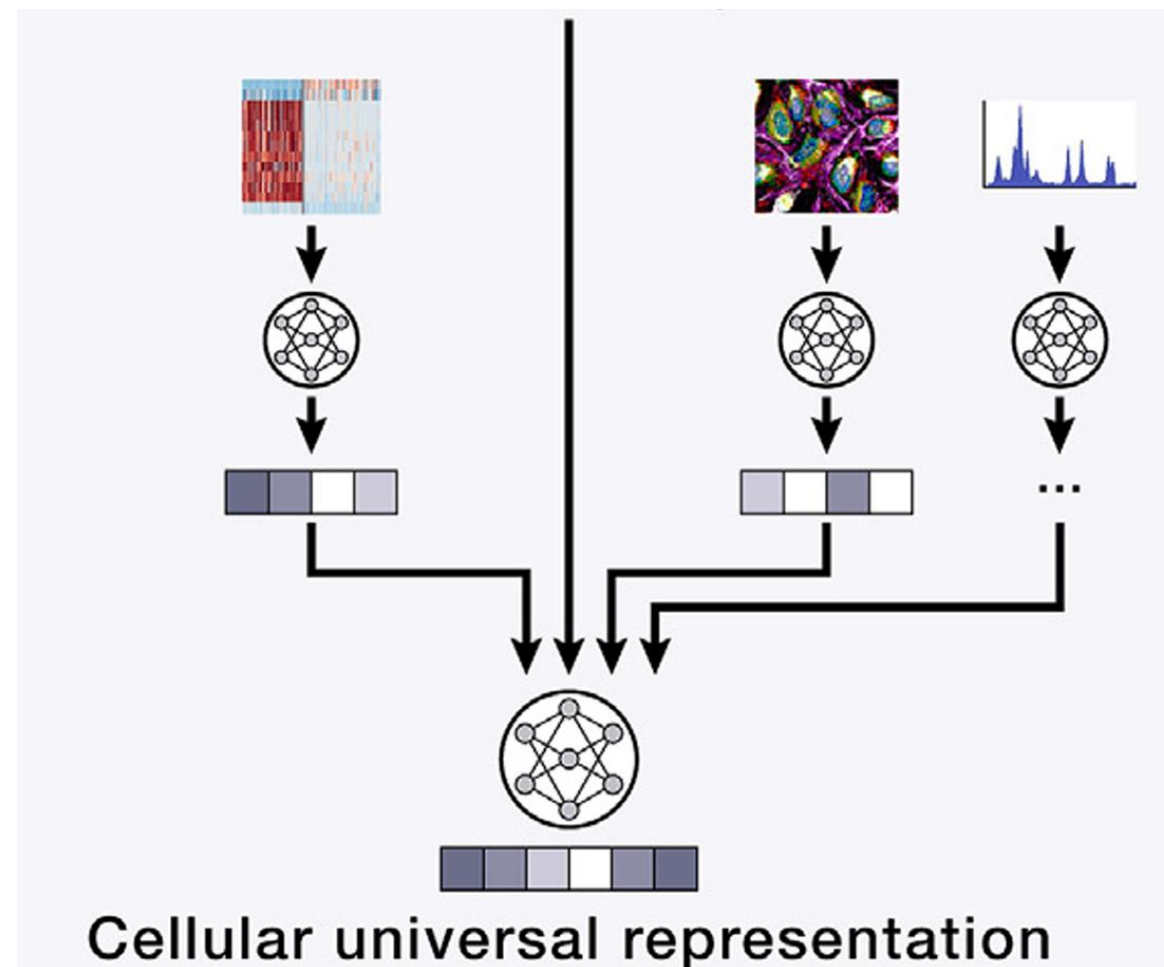
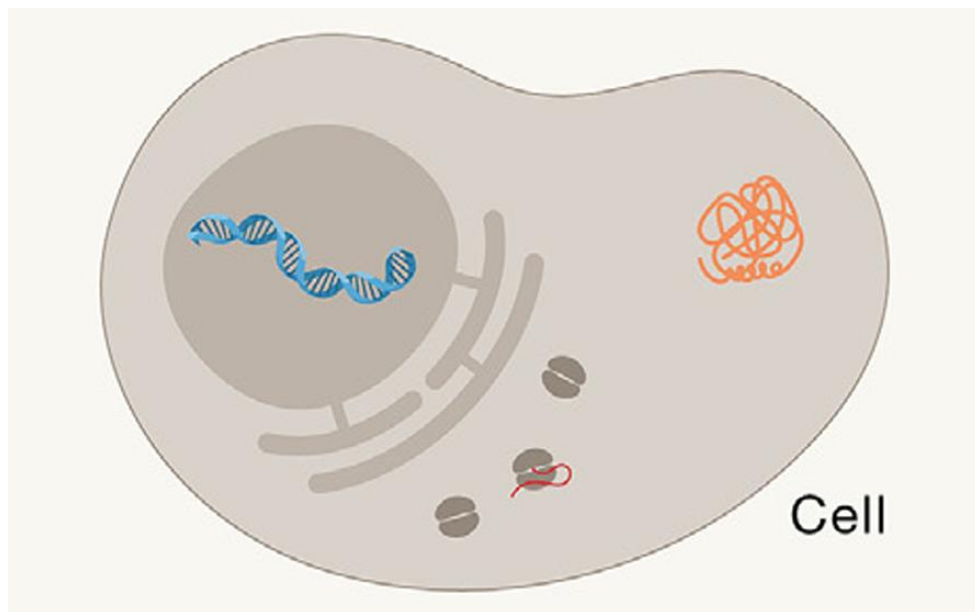
C. Bunne, et al. How to build the virtual cell with artificial intelligence: Priorities and opportunities, Cell 2024



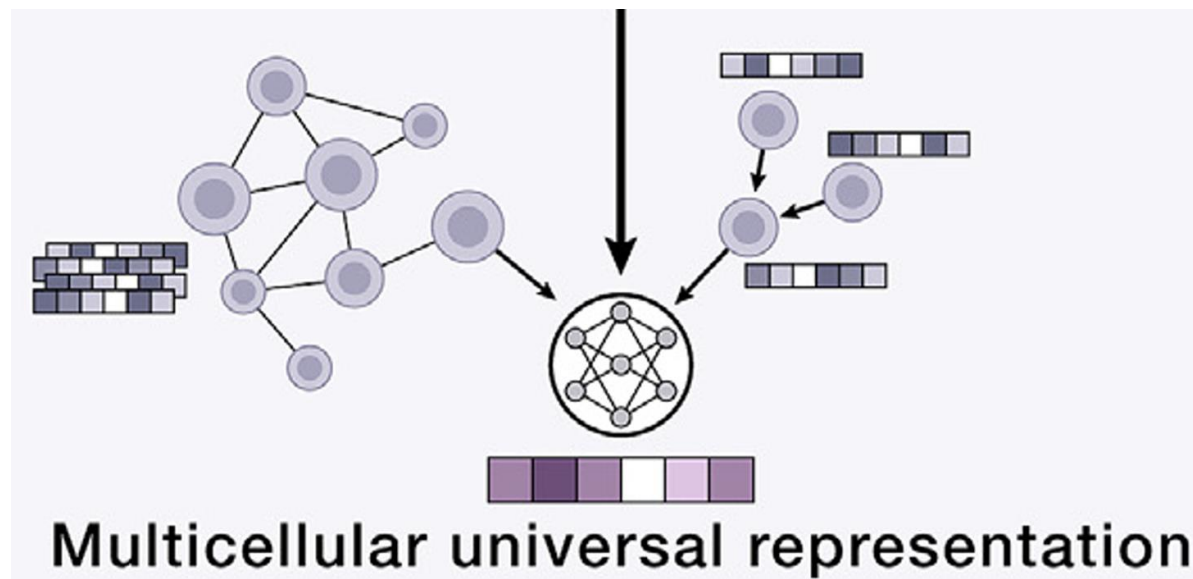
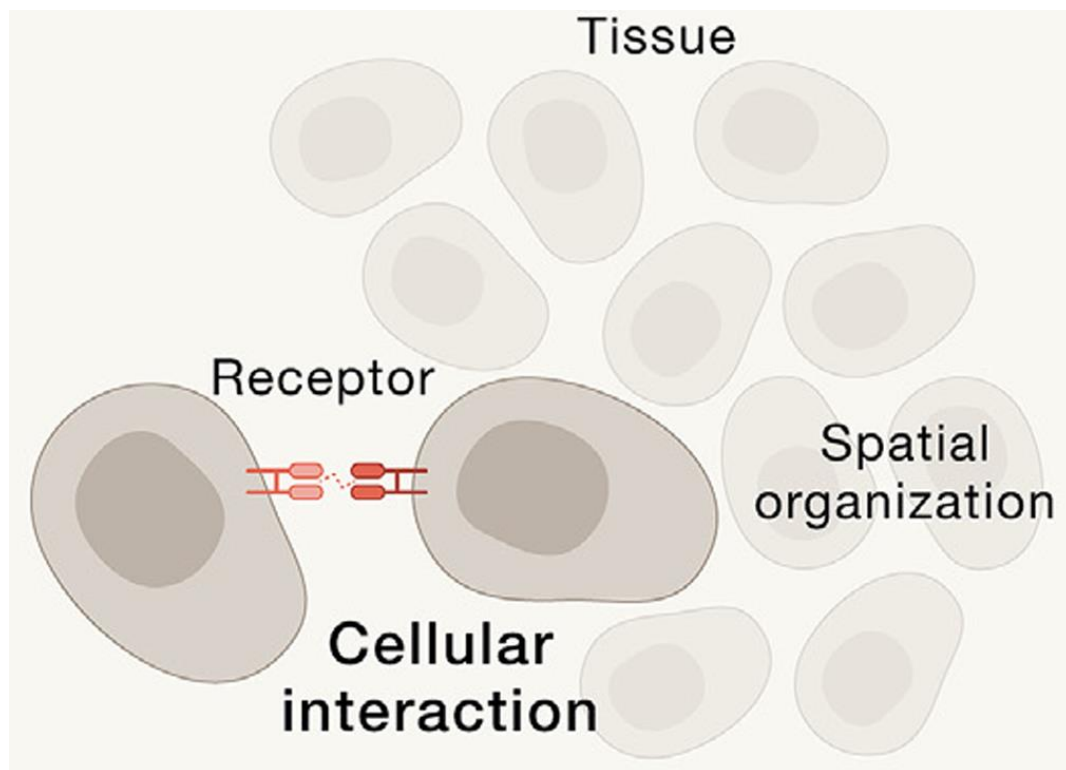
■ UR at molecular scale



UR at cellular scale

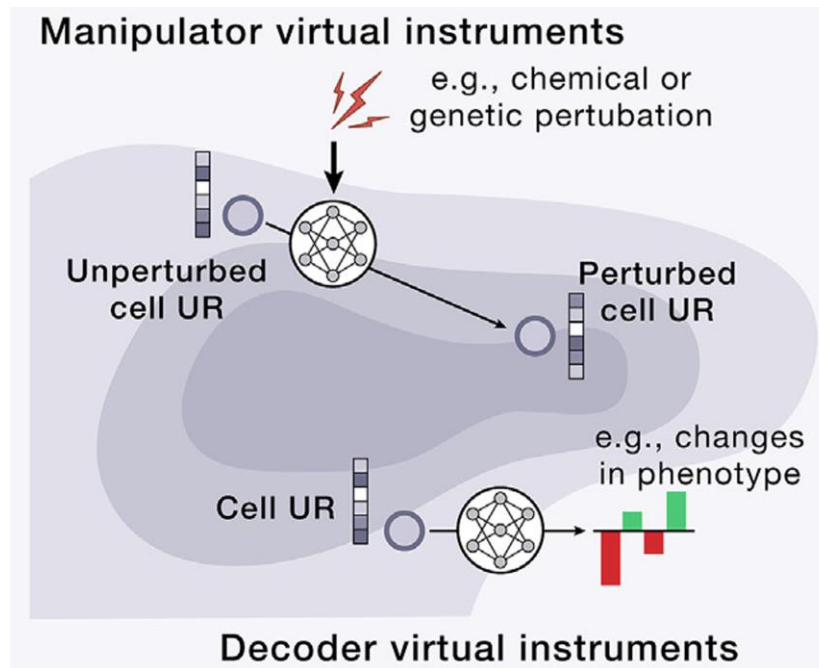
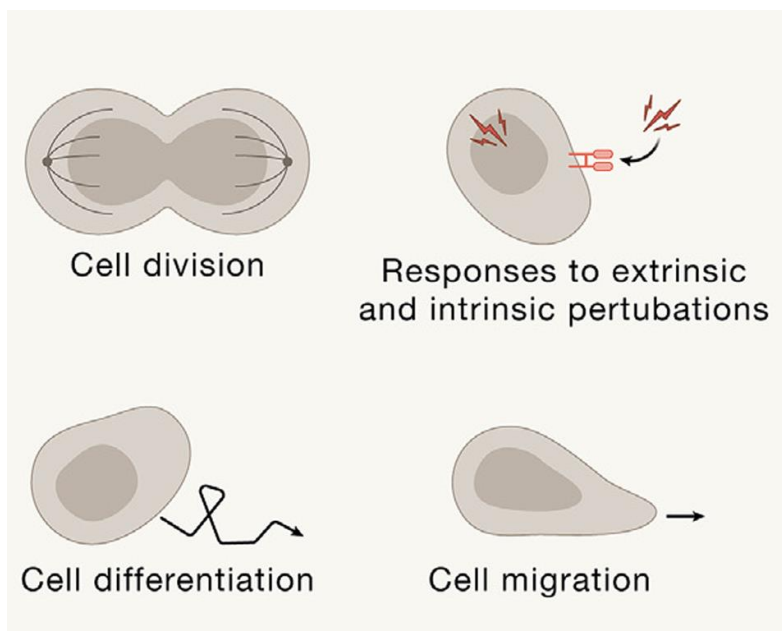


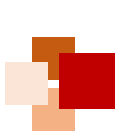
UR at multicellular scale



Virtual instruments (VIs)

- **Virtual instruments (VIs)** are the tools that operate on UR embeddings and perform various functions and tasks
 - **Manipulators:** By altering molecular, cellular and tissue URs, manipulator VIs abstract complex dynamic processes as the transitions between (distributions of) their representations
 - **Decoders:** Take an embedding of biological entities and predict one or more concrete properties, e.g. physical structure, cell type/state, fitness, expression, or drug response

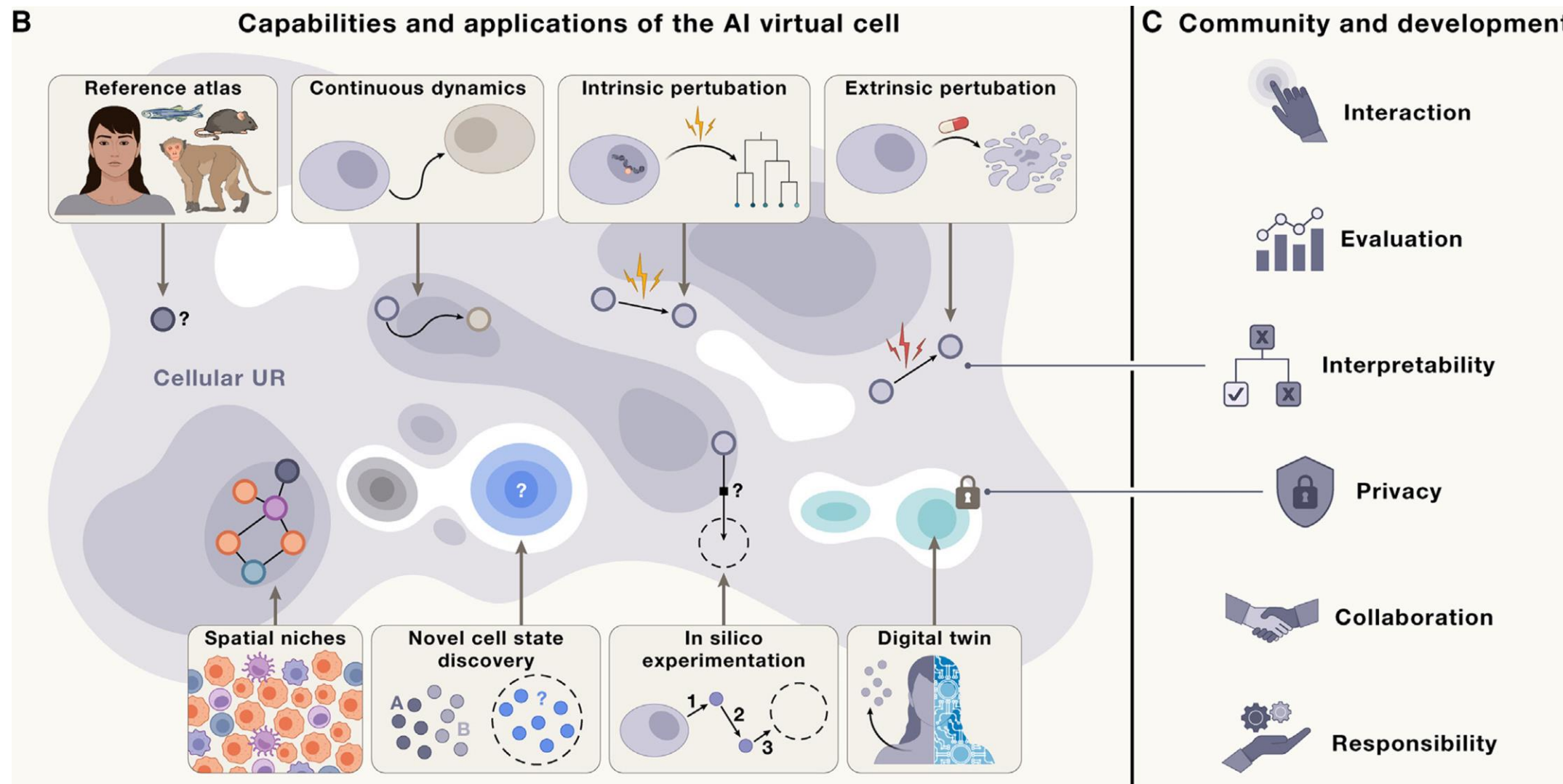




Capabilities and applications of AIVC

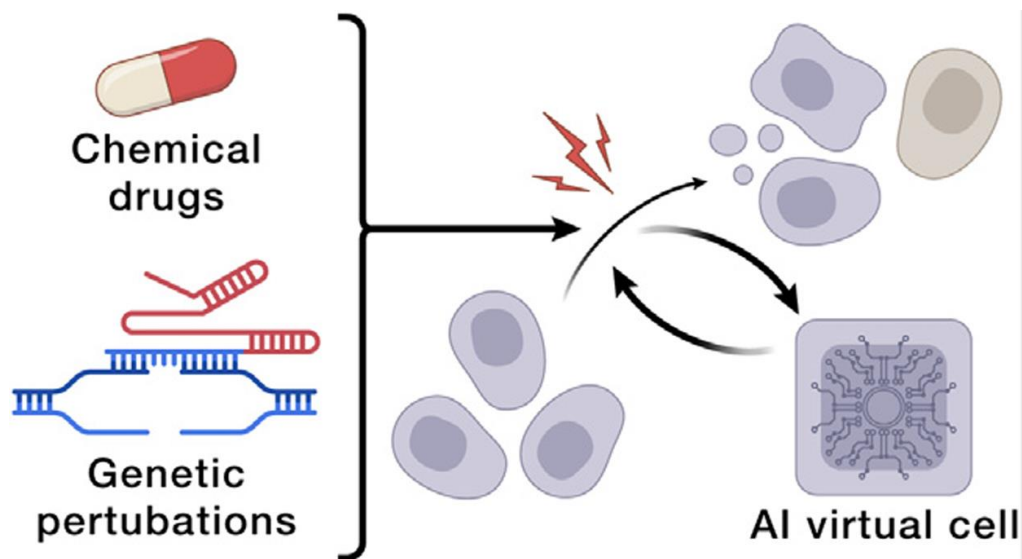


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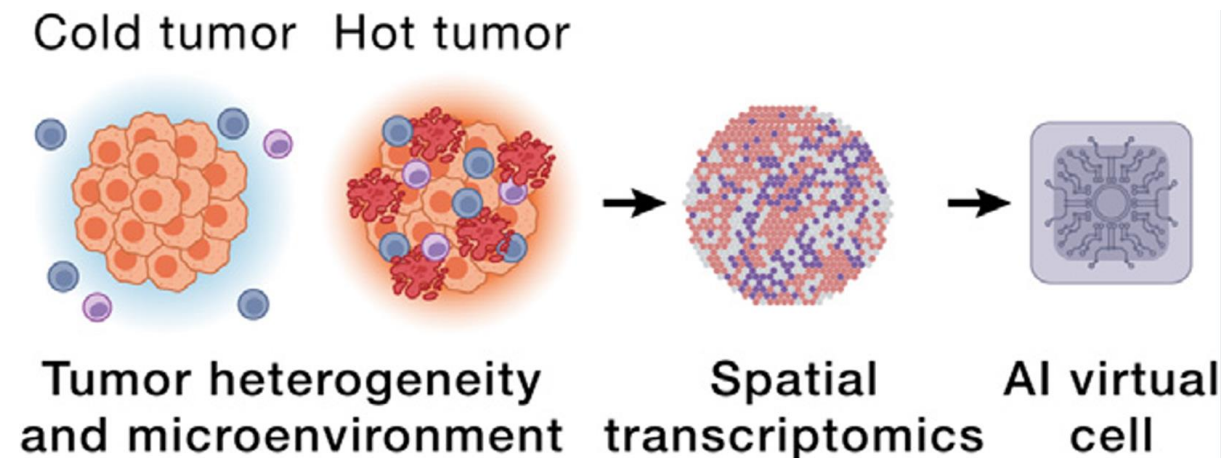


C. Bunne, et al. How to build the virtual cell with artificial intelligence: Priorities and opportunities, Cell 2024

Applications of AIVC



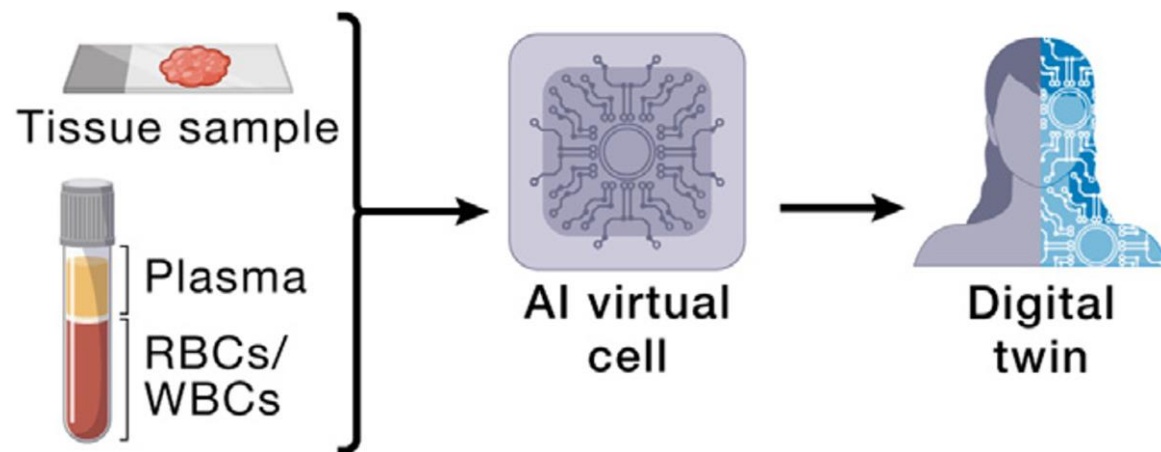
Cell engineering to enable phenotypic drug discovery and cell-based therapeutics



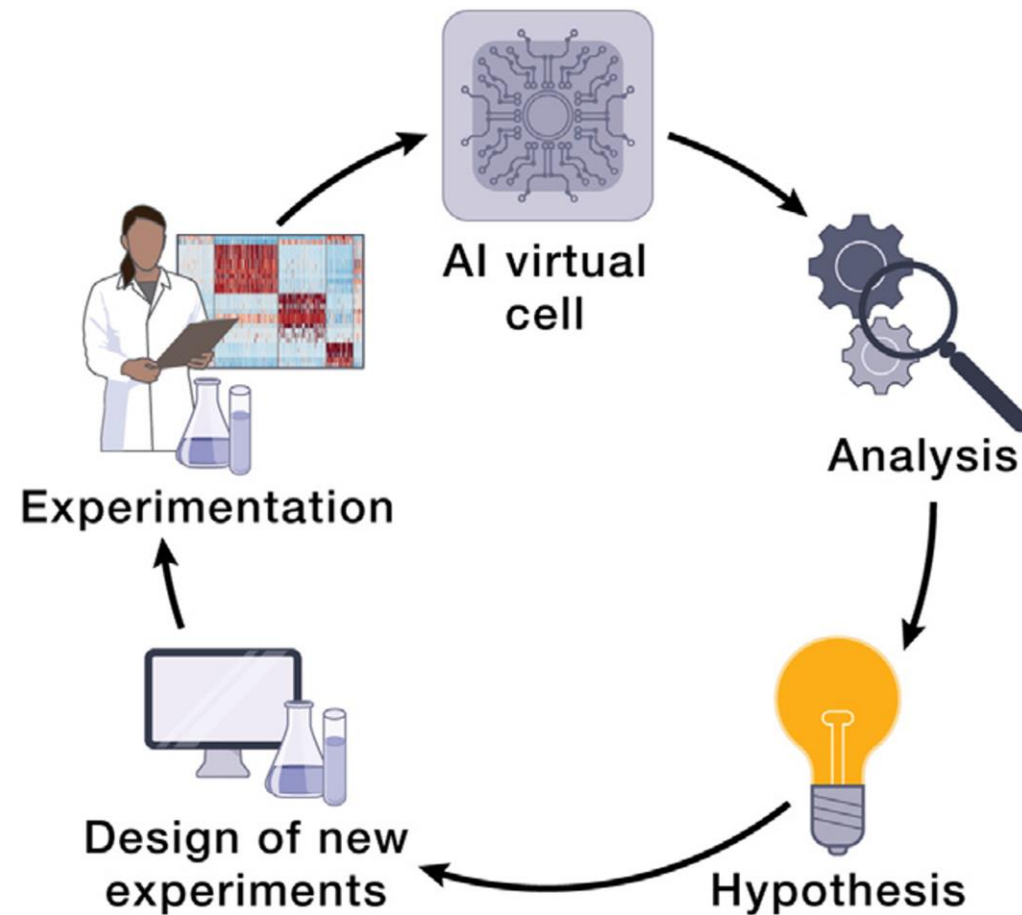
Unlocking the power of spatial biology to fight cancer

C. Bunne, et al. How to build the virtual cell with artificial intelligence: Priorities and opportunities, Cell 2024

Applications of AIVC

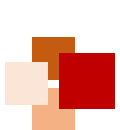


Diagnostic virtual cell models for individual patients



A hypothesis-generating framework for scientific research

C. Bunne, et al. How to build the virtual cell with artificial intelligence: Priorities and opportunities, Cell 2024



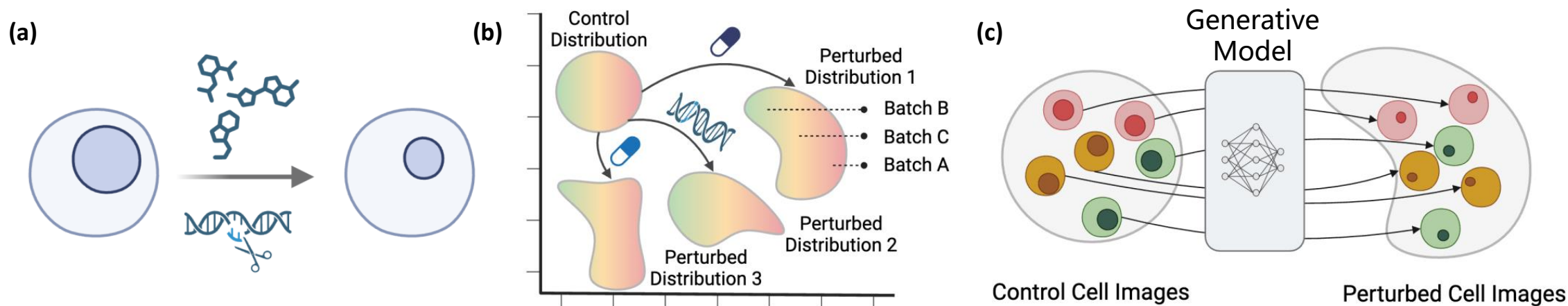
Generative AI for perturbation modeling



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基因扰动和细胞表型变化的关联预测

基于生成式人工智能模拟扰动引起的细胞表型变化



Zhang Y, Su Y, Wang C, et al. CellFlow: Simulating Cellular Morphology Changes via Flow Matching. arXiv, 2025.



- Data needs and requirement
- Model evaluation
- Interpretability and interaction
- Open collaboration as a scientific community





Recap



- In this lecture, we have learned:
 - The basics of whole-cell modeling (aims, methodology, and challenges)
 - A classic ODE model of cell cycle control
 - AI virtual cell (AIVC) modeling learns directly from data (i.e. it is data-driven)
 - Universal representations (URs) at molecular, cellular and multicellular (e.g. tissue and organ) scales are the foundation of AIVC models
 - Virtual instruments (VIs) enable prediction of cell phenotypic changes upon perturbations and explanation of mechanisms
 - Despite an optimistic future, AIVC is faced with several challenges





References



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- Zhang Y, Su Y, Wang C, et al. (2025) CellFlow: Simulating Cellular Morphology Changes via Flow Matching. arXiv.

